Multilayer critical infrastructure network modeling for equitable resilience

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1 Introduction

The frequency and magnitude of extreme events have significantly increased all over the globe during the past two decades (**kermanshah2017robustness**). The performance of ~~transportation~~ critical infrastructure networks can be seriously affected by these extreme events (**ilbeigi2019statistical**) Transportation systems, in particular, are vulnerable to a widespread range of natural events and disruptions such as such as hurricanes, tornadoes, earthquakes, wildfires, traffic incidents, snowing, flooding, infrastructure failures, among others (**bhavathrathan2018algorithm**), (**twumasi-boakye2018resilience**). For example, **ilbeigi2019statistical** ~~has~~ developed a quantifying resilience framework for the 2012 Hurricane Sandy, which ~~has~~ hit urban areas in New York City and ~~h~~was identified as the fourth costliest hurricane in U.S. history. He has evaluated transportation dataset for a period of four weeks (two weeks before the hurricane and two weeks after it) and considering the closeness centrality index, he found out that the New York City transportation network was affected by the hurricane for 135 hours. The proposed framework helps decision makers to identify more vulnerable parts of the network, which are more likely to fail first (**ilbeigi2019statistical**).

Maybe include another citation addressing another CI.

The last 3 sentences of this paragraph which highlight the work done by Ilbeigi (2019) does not quite belong in the opening paragraph. Fine to cite, but there is too much detail about his/her work at this point. Might be better to move this portion to a more relevant location in the Background section.

In order to develop a more reliable framework for quantifying resilience in transportation networks, the focus of the resilience studies should be shifted from transportation network to a more comprehensive view in which all critical infrastructures elements are included. For instance, one of the most vulnerable elements in road transportation systems are bridges which can highly affect robustness of the transportation network as a whole (**zhang2017resiliencebased**). Moreover, it is critical to have a resilient transportation network during and after a disaster, to ensure emergency services, rescue operations, and access to major population and activity centers in a city (**ilbeigi2019statistical**). Although it is very important to make decisions regarding restoration of transportation networks in a short period after event, these decisions have to be made ~~in a way that~~ with regard to various decision-making criteria and resource constraints have been made in to account (**liu2020optimal**). So it is of vital importance for decision makers to have access to a promising decision making framework able of quantifying transportation network resilience. This way, they can ensure availability of a promising transportation network with the optimum capacity, when a disaster occurs.

I like the opening sentences here. Very good.

It has been a long time since researchers started applying network analysis as an effective method in studying the behavior of complex systems, such as social, biological and transportation systems. Representing transportation networks through graphs, is

one of the preliminary steps in quantifying transportation networks, which have been conducted by many resilience studies. There are different approaches in modelling transportation networks from considering cities as nodes and traffic roads as edges (**ip2009resilience**), and considering potential traffic bottlenecks in a city as nodes and streets as edges (**das2020approach**) to considering intersections and state bridges as nodes and interstates, highways or spurs as edges (**antony2017developing**). Traditional network studies fail to provide a comprehensive overview of a complex system by ignoring the presence of multiple type of edges of a system in which entities have a different set of neighbors in each layer (**domenico2013mathematical**). Focusing on multilayer networks in the field of network resilience is very important due to the criticality and complexity related to these infrastructures. Transport process in a multilayer network is more complicated than a single layer network (**wu2020traffic**) and it is very important to reflect these complexity in decision making framework.
Moreover, one of the important criteria, which has been neglected in studies on transportation network's resilience, is social equity. People's vulnerability is affected by their exposure to hazard and their ability to avoid or manage the effects of the hazard (**cutter1995race**). Studies have shown that class, race and ethnicity, gender, disability, political power, personal wealth, age, housing ownership and occupation can affect vulnerability against hazards (**bolin2018race**),(**cutter1995race**). Although there are many studies focusing on considering social equity in planning transportation networks, there is a significant gap in including social equity as an important factor in developing equitable framework for promoting resilience in transportation system's infrastructure.
Considering transportation systems as the main foundation to provide support for national security, economy, public welfare and individuals' daily activities (**qiang2019empirical**), it is critical to develop a comprehensive quantifying framework for network resilience. Although previous studies provide valuable insights about transportation network resilience, there is an urgent need to broaden this concept by considering equity metrics and also by encompassing critical infrastructure systems including bridges, buildings, etc as a multilayered network to develop a decision making framework for restoration of a transportation infrastructure in case of a disaster.

2 Background

Recently, the focus of transport infrastructure management has switched from protection to resilience (**das2020approach**). Moreover, as a result of the developments in resilience engineering studies, the definition of system resilience has shifted from system's ability to deal with threats to system's ability to adjust its functionality (**hollnagel2010resilience**). Based on this new definition, a system is resilient when it can respond to what happens, monitor critical developments, anticipate future threats and sustain required functions (**hollnagel2010resilience**). The resilience of a network system refers to its capability to withstand, adapt to and recover fully or partially after a disruptive event (**aydin2018framework**), (**almoghathawi2019component**), which can be natural or man-made (**ahmed2020resilience**).
weilant2019incorporating have suggested three different aspects associated with the concept of resilience:

- Decreasing the probability of a disaster and increasing the ability of a community to resist a disaster
- Increasing the adaptability of a system while maintaining functions in the presence of a disaster
- Decreasing the needed time for the system to recovery to normal functioning

Ten dimensions of resilience in transportation systems have been identified. These dimensions are redundancy, diversity, efficiency, components' dependency, strength, stakeholders' collaboration (**ahmed2020resilience**), adaptability, mobility performance, safety performance, and the ability to recover quickly (**ahmed2020resilience**), (**murray-tuite2006compa**

3 Overview of relevant resilience methods and approaches

Critical infrastructure are those that if destroyed or rendered unavailable will significantly impact social or economic welfare or affect national security, national public health or safety. (**skrlj2019py3plex**). Having a resilient critical infrastructure is vital for a society in order to resist, respond to, and recover from disruptive events (**tingting2020transportation**). Transportation infrastructure has been identified as one of critical infrastructure systems essential to the well-being of modern societies (**zhang2016resiliencebased**).

Research on transportation system resilience began in the 1990s (**ahmed2020resilience**). Since the concept of resilience engineering has emerged, different related concepts and methods has been addressed by many scholars and most of the studies on transportation system resilience studies focused on the resilience of the roadway-based transportation system (**ahmed2020resilience**). At the beginning of this journey, most of the studies focused on providing qualitative overview of resilience (**baroud2014inherent**), (**bhatia2015network**) but providing a quantitative overview of resilience is attracting more attention during past years and several quantifying framework for transportation networks resilience have been proposed by scholars (**adjetey-bahun2014simulationbaseda**), (**cavallaro2014assessment**), (**bhatia2015network**), (**donovan2017empirically**).

Network system resilience can be modelled considering two phases: pre-disaster phase and post-disaster phase (**ahmed2020resilience**). Some studies have further categorized the post-disaster phase into two sub-phases: response and restoration. The response phase is shortly after the disruptive event and restoration phase follows the response phase. During restoration phase the infrastructure is waiting for repair, but daily travel demand has recovered to a relatively normal level, although some demand cannot be satisfied by the degraded infrastructure network (**tingting2020transportation**). Optimal recovery strategies to a disaster within a given network may differ by hazard, community /cluster/neighborhood or objective (**bhatia2015network**). So it is critical to develop a decision making framework, which is capable of reflecting changes per phases and modifying itself based on the situation. As a result, developing a dynamic model, which considers multiple efficient measures for quantifying resilience and the effect of different kinds of disasters from mild to sever, becomes more and more important.

In order to come up with an efficient model, it is critical to evaluate all relevant historical data. With increasing availability of data, machine learning technique is applying

in many areas. Machine learning is a technique to predict the behaviour of a system based on an analytical model build upon learning from the latest patterns of historical data (**tizghadam2019machine**). One of the most prevalent application of machine learning techniques, is in time prediction based on current and past data. A significant number of studies have focused on predicting future delays in transportation networks (**chandramouleeswaran2018machine**), (**takeichi2017prediction**), (**choi2016prediction**).
Machine learning technique is also applyed in many other areas such as smart transportation, intelligent transportation networks, car sharing placement, public transport locationing (**tizghadam2019machine**), traffic assignment, predicting behavior of an individual driver in various traffic flow conditions, providing driving decision algorithm for self-driving cars and predicting speed in roadways (**bhavsar2017chapter**).
Machine learning techniques have also been employed toward system resilience. For example, there is a study that focus on estimating clearance time after an incident is investigated (**tang2020statistical**). This way, there can be a real-time decision support for incident management, which helps saving lives and reducing incident recovery time (**bhavsar2017chapter**).

3.1 Overview of multilayer networks

Multilayer networks are networks that contain at least two types of nodes or two types of edges (**skrlj2019py3plex**). Multilayer networks are widespread in real world systems such as communication systems, social systems, transportation systems, and biological systems (**wu2020traffic**). As it is shown in Table 1, following different network's components can be considered in modelling multilayer networks:

Table 1 Multilayer network components

Multilayer network item	**References**
Roads	(**antony2017developing**),(**zhang2017resiliencebased**)
Bridges	(**antony2017developing**), (**zhang2017resiliencebased**)
Walkways	(**antony2017developing**)
Bike paths	(**antony2017developing**)
Private roads	(**antony2017developing**)
High speed and low speed as different layers in a multilayer network	(**wu2020traffic**)
Public transit	(**ahmed2020resilience**)
Passenger and freight car transportation systems	(**ahmed2020resilience**)
Railway	(**bhatia2015network**)

There are different approaches in modelling transportation networks as a multilayer network. (**wu2020traffic**) believes that there are two typical multilayer network models (**wu2020traffic**):

- Multilayer model with a logical layer and a physical layer
- Multilayer model with abstraction of real multilayer networks, in which entities can travel across multiple network layers (for example a high speed layer and a low speed layer)

kurant2006layered have proposed a generalized multilayer network model in which the lower layer represents the physical infrastructure and the upper layer represents the

traffic flows and applied it to three transportation systems. This generalized framework facilitates description, comparison and analysis of complex systems such as transportation networks (**kurant2006layered**). **ip2011resilience** have represented transportation networks by an undirected graph with the nodes as cities and edges as traffic roads and calculated the network resilience of a city node by the weighted average number of reliable passageways with all other city nodes in the network (**ip2011resilience**). **aleta2017multilayer** have presented a model for representation of the transportation system of an entire city by considering each line of each mode of transport (bus, metro, and tram) as a single layer and each stop as a node. Based on their multilayer model, there will be a weighted link (based on distance) between two nodes on a layer if the corresponding line passes through both of them. They have also added a new layer which represents the land to introduce the possibility of moving through the city by walking. This model has been applied to study the interconnected structure of 9 different cities in Europe (**aleta2017multilayer**). **li2019modeling** have developed an innovative approach in resilience planning of multilayer transportation networks by simulating the network dynamics of inter-organizational coordination among interdependent infrastructure systems including transportation, flood control, emergency response, community development, and environmental conservation. They have examined their model to 35 organizations from five infrastructure systems within Texas, based on the data gathered from a Hurricane Harvey and evaluated the weaknesses in coordination between entities (**li2019modeling**). **asgari2016ctmapper** proposed an unsupervised mapping algorithm (CT-Mapper) to map sparse cellular trajectories and modeled a transportation network by a large multilayer graph, in which there are three different layers of train, subway and road. In this multilayer model, nodes are metro/train stations or road intersections and edges are connections between them (**asgari2016ctmapper**).

3.2 Modeling approaches

Different methods have been applied in studying the resilience of a transportation system. **ahmed2020resilience** have categorized modeling framework for quantifying transportation resilience studies into two different categories (**ahmed2020resilience**):

- Step-wise assessment (qualitative assessment, simulation)
- optimization

Bhavathrathan and Patil believe that resilience in a system can be quantified by evaluating the magnitude of faults that a network can survive before collapsing from a state of normal operation. Considering this fact, they have presented a weighted fictitious game theory based algorithm for solving the critical state identification problem (**bhavathrathan2018algorithm**).
Serulle et al. have applied fuzzy logic to develop a quantifying resilience framework. Their fuzzy-based model, allows recognizing different values (estimates, lower quality data, linguistically captured data) and assigns a measurement range to each qualitative or quantitative resilience metric and combines them into a resilience index for transportation infrastructure (**serulle2011resiliency**).
Antony (2017) has presented a disaster restoration framework through modelling a road transportation network with bridges as nodes and interstates or highways as edges using graph theory analytic and Eigen-vector centrality applying open source Python

package of OSMnx (**antony2017developing**).
Chen et al(2018) have investigated strategic investments on enhancing transportation network resilience against man-made emergency events applying network game theory approach, through which, the interactions among neighbour players in a pre-hinterland container transportation network have been evaluated (**hong2018strategic**).
Aydin et al (2018) have applied topology-based simulation using graph-based connectivity metric of Giant Connected Component(GCC) to evaluate road recovery strategies in rural transportation networks following geohazards specially earthquakes. They have also used Monte Carlo to simulate recovery time and quantifying the uncertainty during the recovery process to each strategy (**aydin2018framework**).
Tingting and Yu (2020) have developed a bi-objective bi-level optimization framework for transportation system restoration plan to balance measures of unmet demand and travel time and determine optimal resource allocation for roadway restoration (**tingting2020transportation**).

3.3 Summary of network-based resilience metrics

The first step in developing a decision making framework for restoration of transportation networks, is to define relevant resilience metrics in order to reflect the ability of a multilayered network including transportation system and critical infrastructures to respond to and recover from a disaster. Different resilience metrics have been considered in studies conducted on transportation network'resilience. As it is shown in Table 2 These metrics include:

Figure 1, shows transportation network resilience metrics based on six domains of safety-based metrics, restoration process-based metrics, performance-based metrics, equity-based metrics, network-based metrics, and infrastructure-based metrics.

Table 3 provides an overview from all researches which have been reviewed in section 3 and compares them based on applied resilience metrics, case studies, applied method, and provides a brief explanation from their results.

Table 2 Transportation network resilience metrics

Multilayer network item	**References**
The mean recovery time [1]	(**aydin2018framework**) (**ahmed2020resilience**) (**fang2016resiliencebased**) **almoghathawi2019component** (**baroud2014inherent**) (**zhang2017resiliencebased**) (**ilbeigi2019statistical**) (**pant2014static**)
Vulnerability/ reliability/ network robustness/ probability of failure [2]	(**aydin2018framework**) (**ahmed2020resilience**) (**jenelius2006importance**) (**almoghathawi2019component**) (**zhang2016resiliencebased**)
Unmet travel demand	(**ahmed2020resilience**) (**jenelius2006importance**) (**bhatia2015network**) (**kermanshah2017robustness**) (**tingting2020transportation**)
Average transmission time	(**ahmed2020resilience**) (**wu2020traffic**) (**tingting2020transportation**) (**jenelius2006importance**)
Change in network capacity	(**wu2020traffic**) (**ahmed2020resilience**) (**jenelius2006importance**)
Restoration costs	(**aydin2018framework**) (**ahmed2020resilience**) (**baroud2014inherent**)
Components' dependency, Connectivity and accessibility	(**ahmed2020resilience**) (**bhatia2015network**) (**jenelius2006importance**) (**mattsson2015vulnerability**)
Efficiency of the recovery process	(**aydin2018framework**) (**zhang2017resiliencebased**)
loss of service cost	(**fang2016resiliencebased**) (**baroud2014inherent**)
Accessibility to resources	(**aydin2018framework**) (**ahmed2020resilience**)
Safety performance	(**ahmed2020resilience**)
Shortest path	(**ahmed2020resilience**)
Size of population affected by blockages at each node	(**aydin2018framework**)
Road hierarchy	(**aydin2018framework**)
Average transmission speed	(**aleta2017multilayer**)
Travel cost [3]	(**jenelius2006importance**)
Change in the performance of network [4]	(**ilbeigi2019statistical**)
The time averaged level of operability [5]	(**pant2014static**)
Maximum loss of functionality [6]	(**pant2014static**)

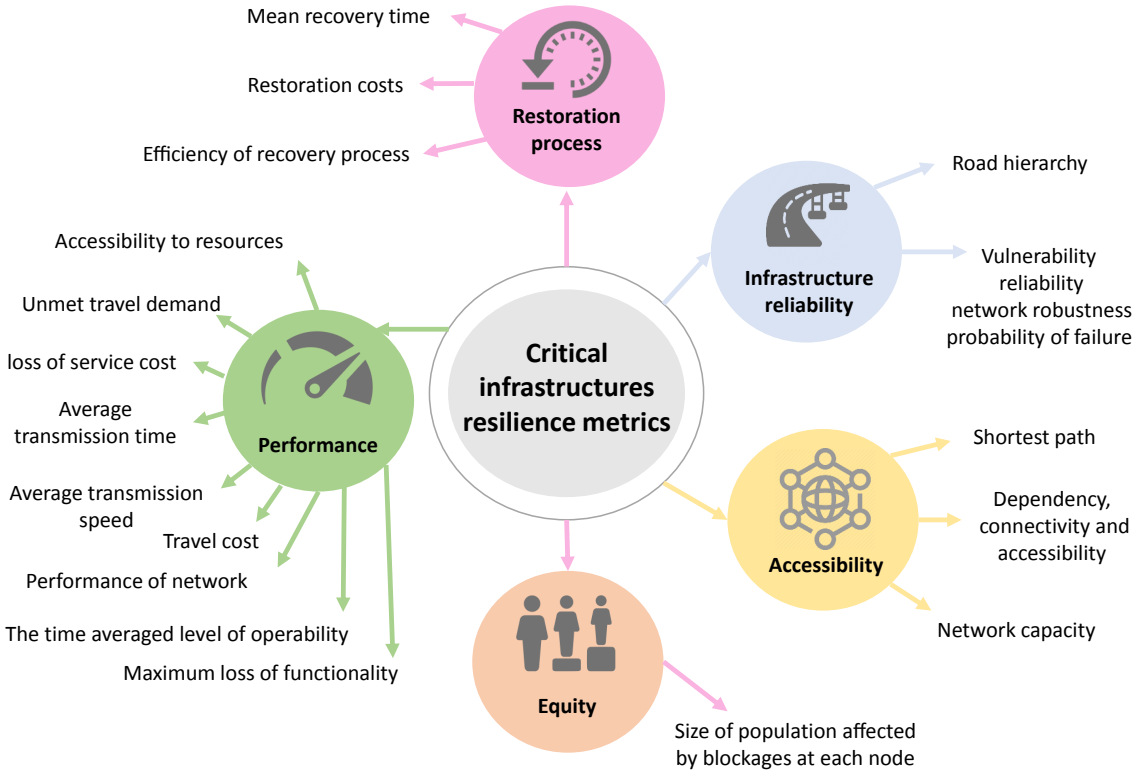

FIGURE 1 Currently used metrics

4 Multilayer network model for an equitably resilient system: a conceptual framework

5 Conclusion

Table 3 Overview of the researches

Reference	Resilience index	Methods	Multi-layer	Disaster	Case study	Results
Serulle et al (2011)	Road available capacity, Road density, Alternate infrastructure proximity, Level of intermodality, Average delay, Average speed reduction, Personal transport cost, Commercial–industrial transport cost, Network management	Fuzzy logic	No	Flooding	Santo Domingo, Dominican Republic	Identifying available capacity, alternate infrastructure proximity, and network management as the most important metrics
Antony (2017)	Centrality measures	Graph theory analytic	No	natural or man-made disaster	Air and road transportation system of the U.S	Identifying nodes with highest Eigen-vector centrality measures as the most important nodes to restore.
Chen et al(2018)	Vulnerability	Network game theory	No	Man-made emergency events (labor strikes)	China's pre-hinterland container transportation network	Providing behavioral analysis of different players in strategic investment decisions
Aydin et al (2018)	proximity to the main research center, time to recovery	Topology-based simulation using graph-based connectivity metric of Giant Connected Component(GCC) and Monte Carlo simulation	No	geohazards specially earthquakes	Sindhupalchok District in Nepal	Providing a framework for evaluating road recovery strategies
Bhavathrathan and Patil (2018)	travel time at an upper envelope of operable disruptions	Game theory	No	-	Anaheim city road network	Developing a quantifying resilience framework based on the magnitude of distruptions
Tingting and Yu (2020)	Unmet demand, Total travel time	bi-objective bi-level optimization framework	No	Hurricane, Earthquake, Flood	Road network in Sioux Falls	Determining optimal resource allocation framework for roadway restoration